Matthew Strong

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Education

Stanford University — Stanford, CA
PhD in Computer Science, Robotics/AI. Advisor: Monroe Kennedy, Jeannette Bohg — *Sep 2023 – present*

University of Colorado Boulder — Boulder, CO
Bachelor of Science in Computer Science, Chinese Minor, 4.00 GPA — *Aug. 2017 – May 2021*

Experience

Robotics/AI PhD Researcher, ARMLab - Stanford University — September 2023 – present
ARMLab, Advised by Monroe Kennedy and Jeannette Bohg — *Stanford, CA*

- Research on safe robot manipulation with 3D priors.
- Building on top of the DenseTact sensor for perception.
- Integrated visual-tactile data into 3D Gaussian Splatting.
- Constructed foundational model for stress tensor and deformation estimation for tactile sensors.

Research Intern, 3D Vision - Applied Intuition — June 2025 – May 2026
Applied Intuition — *Mountain View, CA*

- 4D Pretraining for Autonomy.
- Sole first author accepted to CVPR 2026.

Software Engineer - Customer Experience Platform (CXP) — September 2021 – August 2023
Microsoft — *Redmond, WA*

- AI-based solutions to improve the customer journey experience.
- Designed, developed, and deployed the consent for channel optimization feature for CXP.
- Developed Holdout for A/B test experimentation.

Undergraduate Researcher - CU Boulder SBS Lab — January 2018 – present
Sustainable Buildings and Systems Lab, Advised by Wangda Zuo — *Boulder, CO*

- Automated energy building simulation pipeline for energy outputs of building models in different climate zones.
- Building energy prediction models with Generative Adversarial Networks.
- Core contributor of an open source release of 500+ building models collaboration with NREL and Oak Ridge.

Undergraduate Researcher - HIRO Group — January 2020 – September 2021
HIRO (Human Interaction and RObotics Group) Group, Advised by Alessandro Roncone — *Boulder, CO*

- Novel kinematic calibration algorithm to estimate the pose of a novel sensor unit on a robot manipulator. Validated on a real robot control example.
- Proposed a framework for bridging the gap between avoidance and contact, informed by onboard sensors for collaborative robotics.
- Constructed an effective method for probabilistic fusion of external depth data and onboard proximity data to reduce occlusions in a robot's nearby space.

Publications

Real2Sem2Real: 3D Semantic Augmentations for Robot Learning from Single Human RGB Demonstration

In Submission.

Matthew Strong*, Ankush K. Dhawan*, Julien Poffet, Giuse Pham, Jinho So, Monroe Kennedy III

VisTacAlign: Co-Training Dexterous Policies on Tactile Human and Robot Demonstrations

arXiv 2026.

Julien Poffet, **Matthew Strong**, Ankush Dhawan, Baiyu Shi, Shalika Neelaveni, Yujia Yuan, Zhenan Bao, Monroe Kennedy III

MR. CRABS: Mobile RGB-D Camera-LIDAR Robot for Autonomous Bimanual Manipulation and Sensing

ICRA 2026 Workshop on Act to Sense to Act Better.

Giuse Pham*, **Matthew Strong***, Alex Qiu, Joonwon Kang, Monroe Kennedy III

UniQueR: Unified Query-based Feedforward 3D Reconstruction

ECCV 2026.

Chensheng Peng, Quentin Herau, Jiezhi Yang, Yichen Xie, Yihan Hu, Wenzhao Zheng, **Matthew Strong**, Masayoshi Tomizuka, Wei Zhan

LFG: Learning to Drive is a Free Gift: Large-Scale Label-Free Autonomy Pretraining from Unposed In-The-Wild Videos

CVPR 2026.

Matthew Strong, Wei-Jer Chang, Quentin Herau, Jiezhi Yang, Yihan Hu, Chensheng Peng, Wei Zhan

J-PARSE: Jacobian-based Projection Algorithm for Resolving Singularities Effectively in Inverse Kinematic Control of Serial Manipulators

Conditionally Accepted, T-RO 2026.

Shivani Guptasarma, **Matthew Strong**, Honghao Zhen, Monroe Kennedy III

TensorTouch: Calibration of Tactile Sensors for High Resolution Stress Tensor and Deformation for Dexterous Manipulation

T-RO 2026.

Won-Kyung Do, **Matthew Strong**, Aiden Swann, Boshu Lei, Monroe Kennedy III

SPACeR: Self-Play Anchoring with Centralized Reference Models

ICLR 2026.

Wei-Jer Chang, Akshay Rangesh, Kevin Joseph, **Matthew Strong**, Masayoshi Tomizuka, Yihan Hu, Wei Zhan

DexFruit: Dexterous Manipulation and Gaussian Splatting Inspection of Fruit

RA-L 2025.

Aiden Swann*, Alex Qiu*, **Matthew Strong**, Angelina Zhang, Samuel Morstein, Kai Rayle, Monroe Kennedy III

Next Best Sense: Guiding Vision and Touch with FisherRF for 3D Gaussian Splatting

ICRA 2025. ***Oral.***

Matthew Strong*, Boshu Lei*, Aiden Swann, Wen Jiang, Kostas Daniilidis, Monroe Kennedy III

Touch-GS: Visual-Tactile Supervised 3D Gaussian Splatting

IROS 2024. ***Oral.***

Matthew Strong*, Aiden Swann*, Won Kyung Do, Gadiel Sznaier Camps, Mac Schwager, Monroe Kennedy III

SHIRO: Soft Hierarchical Reinforcement Learning

Arxiv.

Kandai Watanabe*, **Matthew Strong***, Omer Eldar

Individualized Empirical Baselines for Evaluating the Energy Performance of Existing Buildings

Science and Technology for the Built Environment.

Yingli Lou, Yunyang Ye, Yizhi Yang, Wangda Zuo, Gang Wang, **Matthew Strong**, Satish Upadhyaya, Chris Payne

Evaluating Performance of Different Generative Adversarial Networks for Large-Scale Building Power Demand Prediction

Energy and Buildings.

Yunyang Ye, **Matthew Strong**, Yingli Lou, Cary Faulkner, Wangda Zuo, Satish Upadhyaya.

Volumetric Data Fusion of External Depth and Onboard Proximity Data For Occluded Space Reduction

4th Workshop on Proximity Perception in Robotics at IROS 2021.

Matthew Strong*, Caleb Escobedo*, Alessandro Roncone.

Contact Anticipation for Physical Human–Robot Interaction with Robotic Manipulators using Onboard Proximity Sensors

IROS 2021. ***Oral.***

Caleb Escobedo, **Matthew Strong**, Mary West, Ander Aramburu, Alessandro Roncone

Self-Contained Kinematic Calibration of a Novel Whole-Body Artificial Skin for Collaborative Robotics

IROS 2021. ***Oral.***

Kandai Watanabe, **Matthew Strong**, Mary West, Caleb Escobedo, Ander Aramburu, Krishna Chaitanya, Alessandro Roncone.

Enabling Close Proximity Human Robot Collaboration via Distributed, Self-Calibrating Robotic Skin

BS Thesis at CU Boulder. Committee: Alessandro Roncone, Bradley Hayes, Christoffer Heckman

Matthew Strong.

Development of New Baseline Models for U.S. Medium Office Buildings Based on Commercial Buildings Energy Consumption Survey Data

Science and Technology for the Built Environment Volume 26, 2020.

Yunyang Ye, Yingli Lou, **Matthew Strong**, Satish Upadhyaya, Gang Wang, Wangda Zuo.

Service/Leadership

Microsoft BAP Early-in-Career 2023

- One of the founding members of a newly revamped Early-in-Career group.
- Leader of the Dynamics 365 Early-in-Career group
- Planned events, including single-handedly starting a AMA with the Corporate Vice President of the org.

HackCU August 2018 – May 2021

- Led HackCU's tech team, organizing the largest hackathon in the Rocky Mountain region.
- Managed whole tech stack during 500+ person hackathon.
- Developed hacker sites and APIs accessed by thousands of people across the nation and globe.
- Handled 1000+ hacker applications.

Slingshot Mentor and Founding Member June 2020 – September 2021

- Served as a founding member for students from top high schools interested in CS at Slingshot, a **Techstars-backed startup** co-founded by students from top tech companies.
- Organized learning sessions about robotics, research, and machine learning for high school students.
- Onboarded startups out of Y-Combinator, Techstars, Berkeley, Stanford, and more.

Discrete Structures Tutor January 2019 – May 2019

- Mentored student in discrete structures.
- Set homework and test-prep deadlines, and prepared practice problems.

SASE (Society of Asian Scientists and Engineers) Leadership January 2018 – May 2018

- Served as Co-Marketing Director and managed social media pages.

Awards

Stanford Centennial Teaching Assistant Award May 2025

- Received top TA award for teaching the Collaborative Robotics Course at Stanford. Only 20 students in the entire School of Engineering receive this award.

National Science Foundation Graduate Research Fellowship April 2024

- Received prestigious NSF GRFP for my PhD studies at Stanford.

Microsoft Special Stock Award July 2023

- The most prestigious bonus given to employees at Microsoft. Only a few individuals were nominated out of the 200+ employees in my organization.

RSS Inclusion Fellow June 2021

- Accepted as an RSS (Robotics: Science and Systems) Inclusion Fellow, an expenses-paid program for upcoming robotics researchers attending RSS.

Chancellor's Recognition Award May 2021

- An award given to students that maintain a 4.0 GPA throughout all of college.

Active Learning Award April 2021

- Award given to outstanding students in service, professional, and research experiences.

College of Arts and Sciences: Class of 2021 Amazing Student April 2021

- For my minor in Chinese, I received an outstanding student award for the Class of 2021 in the College of Arts of Arts and Sciences (1 of 30 out of 4500+ students).

College of Engineering Research Award April 2021

- Only one other CS student from CU has received this award in its 20+ year history.

CRA Outstanding Undergraduate Researchers 2021 – Honorable Mention December 2020

- The most prestigious award for CS undergraduate researchers in North America.

UROP Research Grant: Null Space Control for Collaborative Robotics August 2020 – August 2021

- Received $1500 research grant from CU Boulder's Undergraduate Research Opportunities Program (UROP) to perform research on null space control.

Sewall Scholar August 2017 – May 2021

- The top merit scholarship at CU Boulder.

Engineering Merit Scholarship August 2017 – May 2021

- Received based on high school academic performance.

BOLD Scholarship August 2017 – May 2021

- Received diversity scholarship based on high school achievement.

National Merit Scholar August 2017

- Selected to receive National Merit Scholarship on basis of outstanding high school achievement.
- Given to <1% of high school seniors in the US.

Membership

Microsoft BAP EIC Group March 2022– August 2023

Colorado Data Science Team August 2019 – December 2020

SHPE (Society of Hispanic Professional Engineers) August 2019 – May 2021

SASE (Society of Asian Scientists and Engineers) August 2017 – August 2019

Invited Talks

HackCU Workshop Invited Speaker: An Introduction to AI March 2024

- Gave a talk on an introduction to AI at HackCU.

HackCU Workshop Invited Speaker: A Gentler Introduction to Robotics March 2022

- Invited to conduct a workshop on how to get started with algorithm robotics at HackCU.

Invited Speaker at the PhD Forum: 4th Workshop on Proximity Perception in Robotics August 2021

- Only undergraduate invited to speak at the forum.

Slingshot Interview Series: Getting Involved in Robotics August 2021

- Invited by Techstars-backed startup to be interviewed on my experience with robotics, how I improved, and what others can learn from me.

Reviewer

IEEE International Conference on Robotics and Automation (ICRA) 2023, 2025, 2026, 2027

IEEE Robotics and Automation Letters (RA-L) 2023, 2024, 2025, 2026

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022, 2024

Conference on Robot Learning (CoRL) 2026